

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 24

LCM

AIM: To write and execute an 8085 assembly language program to find LCM of two numbers in an array and store result in memory.

ALGORITHM

1. Store the first number at memory location **2050H**.
2. Store the second number at memory location **2051H**.
3. Load the first number into register **B** and the second number into register **C**.
4. Copy the first number into register **D**.
5. Subtract the second number from the value in **D**.
6. If the result is zero, the value in **D** is the LCM.
7. If the result is negative, add the first number to **D**.
8. Repeat the process until the result becomes zero.
9. Store the LCM at memory location **2052H**.
10. Stop the program.

CODE:

```
LDA 2050H
```

```
MOV B,A
```

```
LDA 2051H
```

```
MOV C,A
```

```
MOV D,B
```

```
LOOP: MOV A,D
```

```
    SUB C
```

```
    JZ DONE
```

```
    JC ADD
```

```
    MOV D,A
```

JMP LOOP

ADD: MOV A,D

ADD B

MOV D,A

JMP LOOP

DONE: MOV A,D

STA 2052H

HLT

OUTPUT:

